MODULE 4: STATISTICAL THINKING (09 PERIODS)

Course Code: 22DS102006 | **B.Tech Computer Science & Engineering (Data Science)**

Detailed 10-Page Comprehensive Study Notes: Histograms, Outliers, Central Tendency, Dispersion, PMF, CDF, Percentiles, Random Variables & Theoretical Distributions.

1. Introduction to Statistical Thinking & Histograms

Statistical thinking is the philosophy of learning from data, quantifying uncertainty, and making rigorous decisions under stochastic environments. Rather than viewing data as static numbers, statistical thinking treats observations as random samples drawn from underlying probabilistic populations.

The first step in exploring empirical data is visualization through **Histograms**. A histogram divides the data range into contiguous, non-overlapping intervals called bins and counts the number of observations falling into each bin.

Key Analytical Insights from Histograms

- **Mode Identification:** Unimodal, bimodal, or multimodal shapes reveal distinct underlying subpopulations.
- **Symmetry vs. Skewness:** Asymmetry indicates long tails (left-skewed or right-skewed distributions).
- **Bin Width Sensitivity:** Excessively narrow bins reveal random noise; overly wide bins mask critical structural peaks.

Empirical Distributions vs. Theoretical Models

Empirical distributions represent raw measurements captured in the wild. Statistical thinking bridges raw counts to mathematical models, allowing data scientists to simulate, predict, and infer properties of populations from finite samples.

2. Outliers and Data Cleaning Principles

Outliers are extreme data points that diverge significantly from the remaining observations in a dataset. They can arise from genuine rare events (e.g., fraudulent financial transactions) or from measurement/data-entry errors.

Detection Mechanisms

- **Interquartile Range (IQR) Method:** Outliers are defined as values falling below `Q1 - 1.5 × IQR` or above `Q3 + 1.5 × IQR`, where `IQR = Q3 - Q1`.
- **Z-Score Method:** Measures how many standard deviations `σ` a data point `x` is away from the mean `μ`: `Z = (x - μ) / σ`. Values with `|Z| > 3.0` are typically flagged.
- **Modified Z-Score:** Uses median and Median Absolute Deviation (MAD), making it robust against contamination by extreme outliers.

> **Handling Strategy Considerations**
>
> Never discard outliers automatically. Investigate their origin: if they represent legitimate extreme phenomena (fat-tailed phenomena), models must account for them; if they represent sensor corruption or typos, truncate, winsorize, or impute them appropriately.

3. Measures of Central Tendency & Dispersion

Summarizing numerical datasets requires metrics that capture both the central location of the data and its spread (dispersion).

Central Tendency Metrics

- **Mean (Arithmetic Average):** $\bar{x} = (1/n) \sum x_i$. Highly sensitive to extreme outliers and skewed tails.
- **Median:** The middle value when data is sorted in ascending order. Robust against extreme outliers.
- **Mode:** The most frequently occurring value in the dataset. Useful for categorical and discrete distributions.

Measures of Dispersion

- **Variance (s^2) & Standard Deviation (s):** Quantifies average squared distance from the mean: $s^2 = [1/(n-1)] \sum (x_i - \bar{x})^2$.
- **Mean Absolute Deviation (MAD):** Average distance from the mean or median: $MAD = (1/n) \sum |x_i - \tilde{x}|$.

> **Choosing the Right Summary Statistics**
> For symmetric distributions, use Mean and Standard Deviation. For skewed distributions (e.g., income, website traffic), use Median and Interquartile Range (IQR) to avoid distortion.

4. Effective Reporting of Results & Effect Sizes

Presenting statistical findings requires clarity, precision, and transparency regarding uncertainty. Misleading summaries or omitting sample sizes distorts scientific conclusions.

Core Guidelines for Statistical Reporting

- **Always Report Uncertainty:** Accompany point estimates with confidence intervals or standard errors rather than isolated numbers.
- **Distinguish Statistical vs. Practical Significance:** A large sample size can yield a statistically significant p-value (`p < 0.05`) for a trivial effect. Always report effect sizes.
- **Effect Size Metrics:** Utilize standardized measures such as Cohen's `d` for group mean differences: `d = (x̄₁ - x̄₂) / s_pooled`.

> **Data Visualization in Reporting**
>
> Supplement numerical tables with box plots, violin plots, or error bar charts. Visual representations prevent readers from missing bimodal distributions hidden behind single summary statistics.

5. Probability Mass Functions (PMF)

A Probability Mass Function (PMF) is a function that gives the probability that a discrete random variable is exactly equal to some value: $P(X = x)$.

Properties of Valid PMFs

- **Non-Negativity:** The probability of any individual outcome must be between 0 and 1: $0 \leq P(X = x) \leq 1$.
- **Total Probability Sum:** The sum of probabilities across all possible values in the sample space S must equal exactly 1: $\sum_{x \in S} P(X = x) = 1$.

```
import numpy as np import matplotlib.pyplot as plt import seaborn as sns # Calculating empirical PMF from discrete observations data = np.random.choice([1, 2, 3, 4, 5], size=1000, p=[0.1, 0.2, 0.4, 0.2, 0.1]) values, counts = np.unique(data, return_counts=True) pmf = counts / len(data) # Plotting PMF plt.figure(figsize=(6, 3.5)) plt.bar(values, pmf, color='#2563eb', alpha=0.8) plt.xlabel("Outcome (x)") plt.ylabel("Probability P(X = x)") plt.title("Empirical Probability Mass Function")
```

6. Cumulative Distribution Functions (CDF)

While PMFs show the probability of specific discrete values, the Cumulative Distribution Function (CDF) maps the probability that a random variable X takes a value less than or equal to x: $F(x) = P(X \leq x)$.

Key Properties & Advantages over PMFs

- **Monotonically Non-Decreasing:** CDF curves always rise or stay flat; $F(x_1) \leq F(x_2)$ whenever $x_1 \leq x_2$.
- **Asymptotic Limits:** $\lim_{x \to -\infty} F(x) = 0$ and $\lim_{x \to +\infty} F(x) = 1$.
- **Noise Reduction:** CDFs smooth out sampling noise inherent in empirical histograms, making visual comparisons between two distributions much cleaner.

> **Calculating Percentiles from CDFs**
>
> The CDF allows direct extraction of percentiles. To find the 50th percentile (median), locate the value x where $F(x) = 0.50$ on the vertical axis and read the corresponding value on the horizontal axis.

7. Percentile-Based Statistics & Random Number Generation

Percentiles partition a dataset into 100 equal groups. The `p`-th percentile is the value below which `p` percent of the observations fall.

Percentiles and Quartiles

- **Interquartile Range (IQR):** Defined as `Q3 - Q1` (the 75th percentile minus the 25th percentile).
- **CDF Inversion for Random Sampling:** To generate random numbers following a target distribution, one generates a uniform random number `u ∈ [0, 1]` and applies the inverse CDF (quantile function): `x = F^-1(u)`.

```
      import numpy as np # 1. Percentile calculations data = np.random.randn(10000) p25, p50, p75 = np.percentile(data, [25, 50, 75]) iqr = p75 - p25 print(f"Q1: {p25:.2f}, Median: {p50:.2f}, Q3: {p75:.2f}, IQR: {iqr:.2f}") # 2. Random number generation using Inverse CDF (Exponential) u = np.random.uniform(0, 1, size=1000) lam = 1.5 exponential_samples = -np.log(1 - u) / lam
```

8. Continuous Theoretical Distributions

Theoretical probability distributions model real-world phenomena mathematically, enabling statistical inference and hypothesis testing.

Exponential, Normal, and Lognormal Distributions

- **Exponential Distribution:** Models time between independent Poisson point processes (e.g., customer arrivals, server failures). PDF: $f(x; \lambda) = \lambda e^{-\lambda x}$ for $x \geq 0$. It exhibits the memoryless property.
- **Normal (Gaussian) Distribution:** The cornerstone of parametric statistics governed by mean μ and variance σ^2. Governed by the Central Limit Theorem.
- **Lognormal Distribution:** A variable whose logarithm is normally distributed. Models multiplicative growth processes and right-skewed economic data.

> **The Central Limit Theorem (CLT)**
>
> The CLT states that the sum (or average) of a large number of independent and identically distributed random variables approaches a normal distribution, regardless of the underlying population distribution.

9. The Class Size Paradox & Sampling Bias

The Class Size Paradox is a classic statistical phenomenon demonstrating how sampling bias can distort perceived averages if observations are sampled proportional to size rather than uniformly at random.

Mathematical Explanation

Suppose a university has 1 class of 100 students and 9 classes of 10 students each. Total students = `100 + 9(10) = 190`.

- **Administrator's Average Class Size:** `(100 + 9×10) / 10 = 19` students per class.
- **Student's Average Experience:** 100 students are in the huge class of 100; 90 students are in classes of 10. Average class size experienced by a random student = `(100×100 + 90×10) / 190 = 10900 / 190 ≈ 57.37` students!

> **Key Data Science Takeaway**
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> Always verify the sampling frame. If observations are weighted by capacity, volume, or connectivity (like social network node degrees), unadjusted sample means will be severely biased upward.

10. Chapter Review & Exam Practice Questions

This concluding section reviews key mathematical concepts from Module 4 and provides practical self-assessment questions for examination preparation.

```
      # Comprehensive Statistical Summary Function import numpy as np def
statistical_summary(arr): q25, q50, q75 = np.percentile(arr, [25, 50, 75]) return { 'mean':
np.mean(arr), 'std': np.std(arr, ddof=1), 'median': q50, 'iqr': q75 - q25,
'skewness_indicator': np.mean(arr) - q50 } sample_data = np.random.exponential(scale=2.0,
size=5000) print(statistical_summary(sample_data))
```

Module 4 Exam Review Questions

- Explain why the Median and IQR are preferred over Mean and Standard Deviation when analyzing highly skewed income data.
- Derive why the Cumulative Distribution Function (CDF) is less susceptible to sampling noise than a standard Histogram.
- Describe the Class Size Paradox in your own words and provide a real-world example outside of classrooms.